

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

1. Coin Toss – Maximum Likelihood Estimation (MLE)

Given:

- Total number of coin tosses (**n**) = 10
- Number of Heads obtained (**k**) = 7

Formula:

$$\hat{p} = k / n$$

Calculation:

Maximum Likelihood Estimation (MLE) estimates the probability of an event by dividing the number of successful outcomes by the total number of observations.

Substituting the given values,

$$\hat{p} = 7 / 10$$

$$\hat{p} = 0.7$$

Interpretation:

The estimated probability of obtaining a Head from the given experimental data is **0.7**. This means that according to the observed sample, the coin produced Heads in **70%** of the tosses.

Answer:

Estimated Probability ($\hat{p}$) = 0.7

2. Student Exam – Maximum Likelihood Estimation (MLE)

Given:

- Total number of students (**n**) = 50
- Number of students who passed (**k**) = 42

Formula:

$$\hat{p} = k / n$$

Calculation:

According to the Maximum Likelihood Estimation method, the probability of passing is calculated by dividing the number of students who passed by the total number of students.

Substituting the values,

$$\hat{p} = 42 / 50$$

$$\hat{p} = 0.84$$

Interpretation:

The estimated probability that a student passes the examination is **0.84**, which means that approximately **84%** of the students in the sample passed the examination.

Answer:

Estimated Probability ($\hat{p}$) = 0.84

3. Defective Light Bulbs – Maximum Likelihood Estimation (MLE)

Given:

- Total number of light bulbs tested (**n**) = 100
- Number of defective bulbs (**k**) = 8

Formula:

$$\hat{p} = k / n$$

Calculation:

MLE estimates the probability of a bulb being defective using the ratio of defective bulbs to the total bulbs tested.

Substituting the given values,

$$\hat{p} = 8 / 100$$

$$\hat{p} = 0.08$$

Interpretation:

The estimated probability that a randomly selected light bulb is defective is **0.08**, which means that approximately **8%** of the bulbs are defective according to the observed sample.

Answer:

Estimated Probability ($\hat{p}$) = 0.08

4. Biased Die – Maximum Likelihood Estimation (MLE)

Given:

- Total number of die rolls (**n**) = 60
- Number of times the face '6' appeared (**k**) = 18

Formula:

$$\hat{p} = k / n$$

Calculation:

To estimate the probability of rolling a 6, divide the number of successful outcomes by the total number of rolls.

Substituting the values,

$$\hat{p} = 18 / 60$$

$$\hat{p} = 0.3$$

Interpretation:

The estimated probability of obtaining the number **6** on the die is **0.3**, indicating that the number 6 appeared in **30%** of all rolls.

Answer:

Estimated Probability ($\hat{p}$) = 0.3

5. Weight Update Using Gradient Descent

Given:

- Input (x) = 4
- Initial Weight (w) = 2
- Actual Output (y) = 10
- Learning Rate (α) = 0.01
- Gradient = 4

Step 1: Calculate the Predicted Output

Formula:

Predicted Output = Weight $\times$ Input

Calculation:

Substitute the given values into the formula.

$$y_{\text{pred}} = 2 \times 4$$

$$y_{\text{pred}} = 8$$

Answer:

Predicted Output = 8

Step 2: Calculate the Error

Formula:

Error = Actual Output – Predicted Output

Calculation:

Substitute the actual and predicted values.

$$\text{Error} = 10 - 8$$

$$\text{Error} = 2$$

Answer:

$$\text{Error} = 2$$

Step 3: Update the Weight

Formula:

$$\text{New Weight} = \text{Old Weight} - (\text{Learning Rate} \times \text{Gradient})$$

Calculation:

Substitute the given values into the Gradient Descent update rule.

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

Answer:

$$\text{Updated Weight} = 1.96$$

Code of Conduct:

I **T Bhupal/192425243** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **bhupalreddy**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation